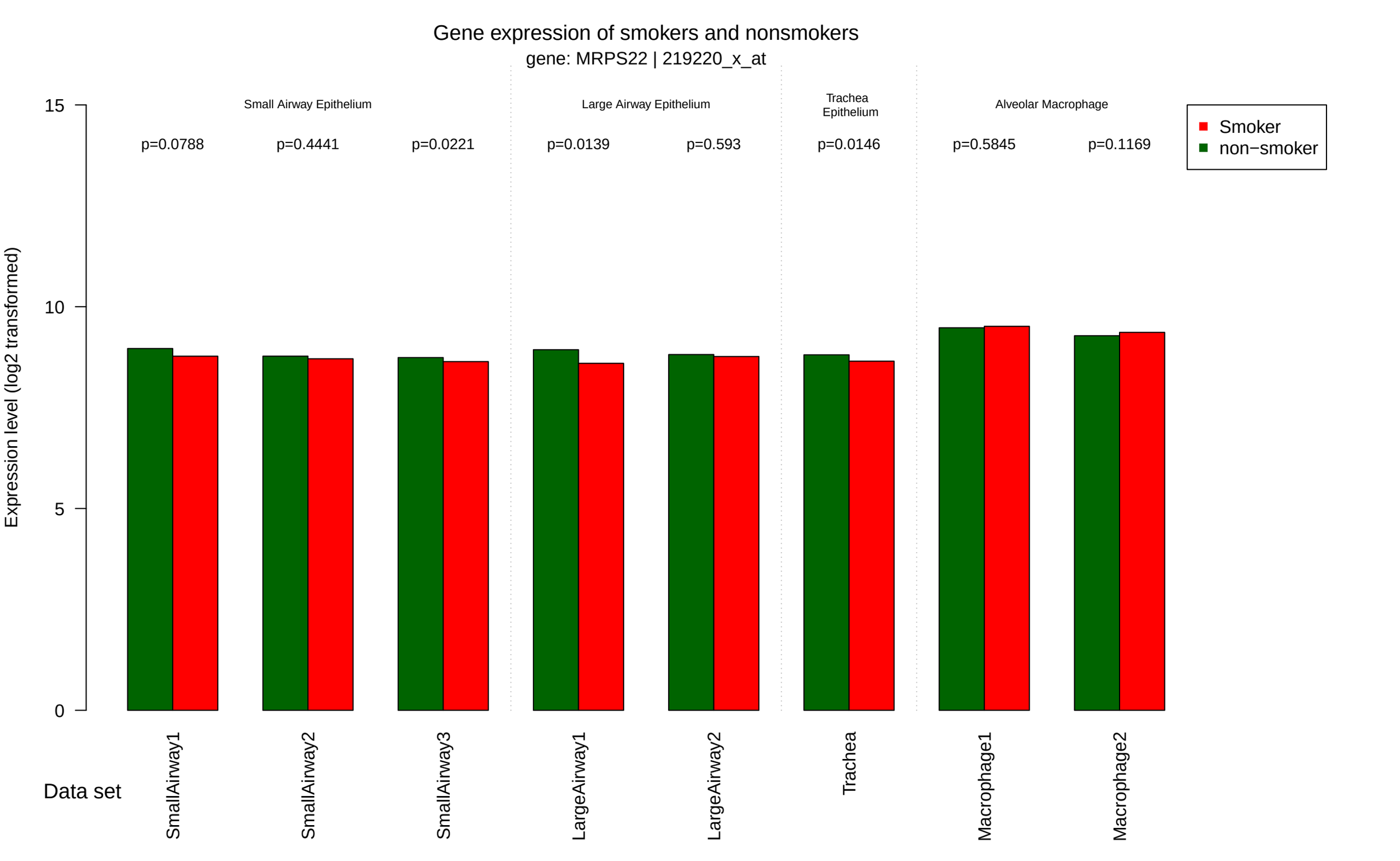
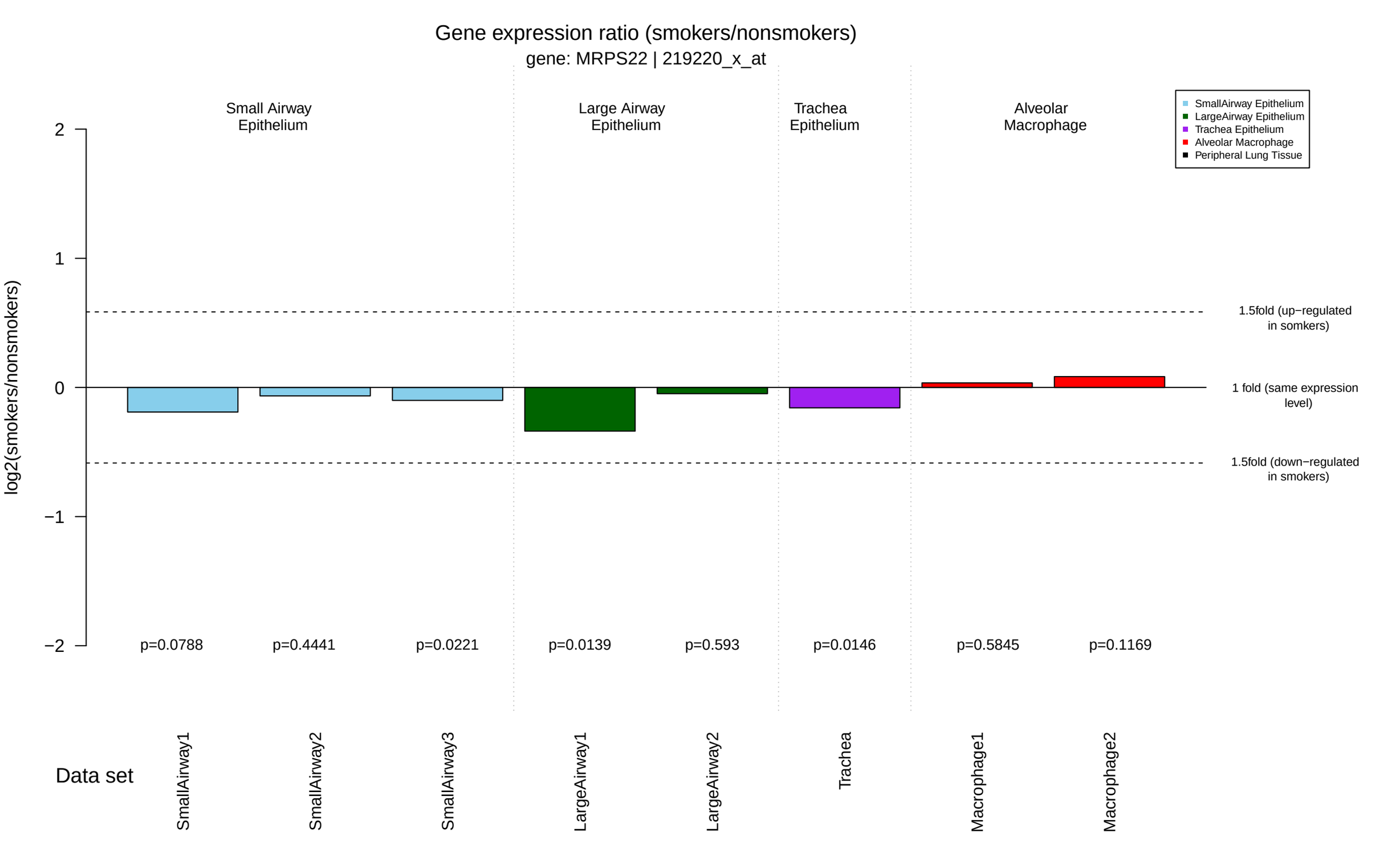
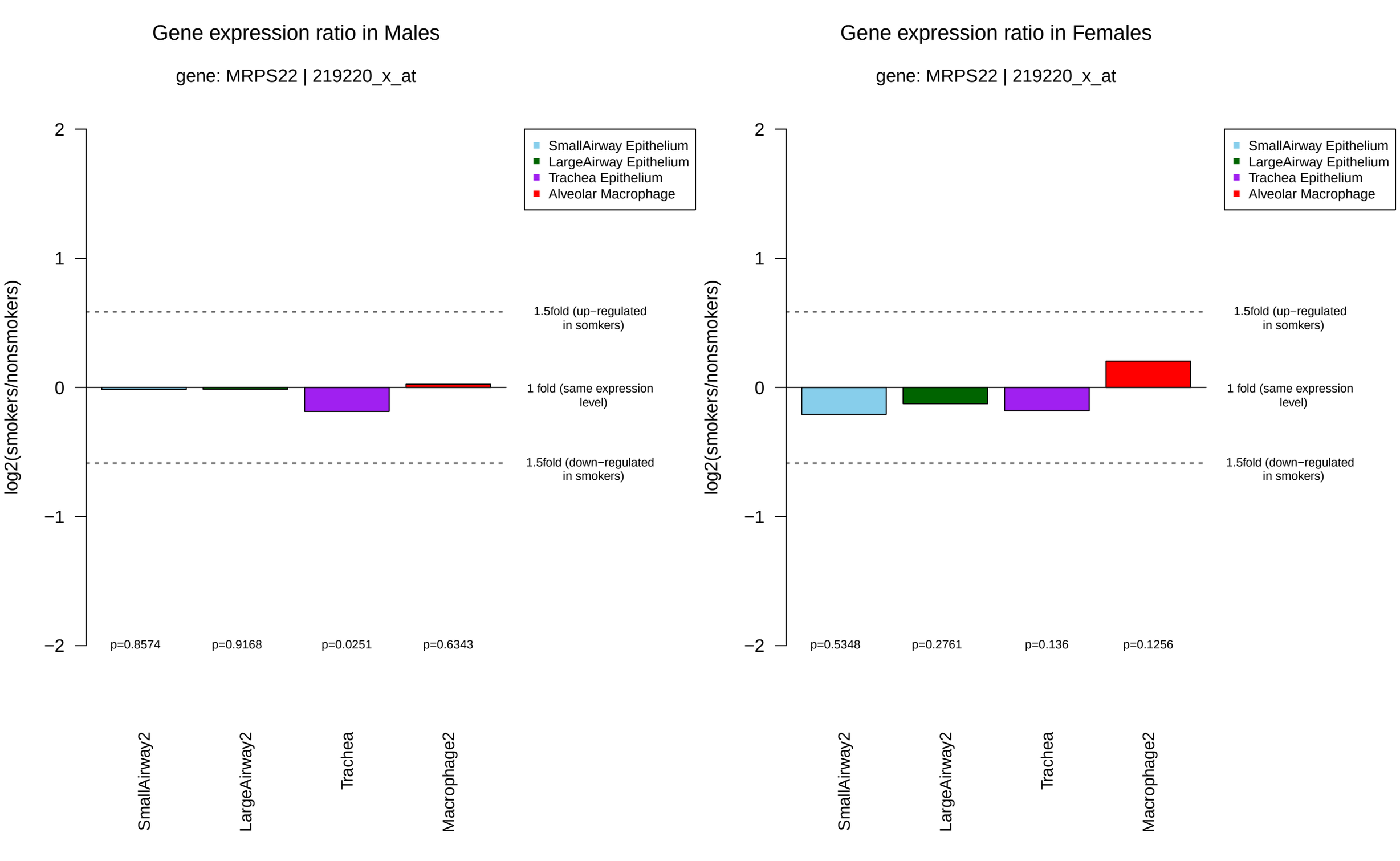
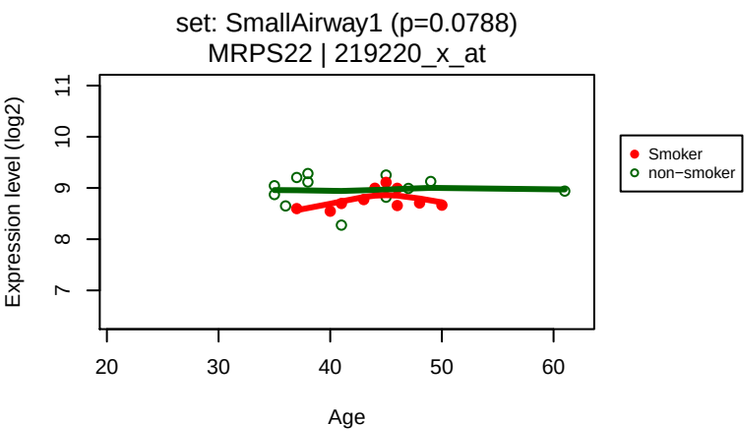
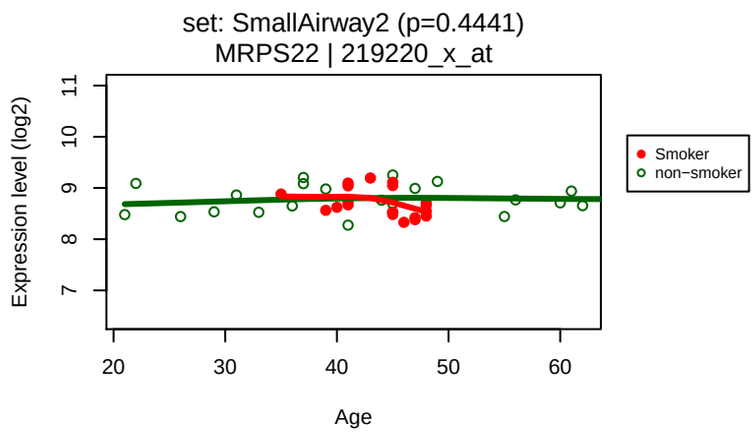
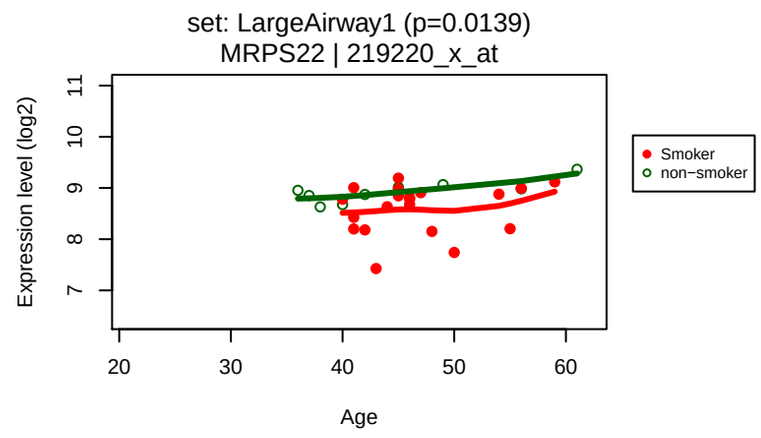

Gene expression of smokers and nonsmokers
gene: MRPS22 | 219220_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0788
p=0.4441
p=0.0221
p=0.0139
p=0.593
p=0.0146
p=0.5845
p=0.1169
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

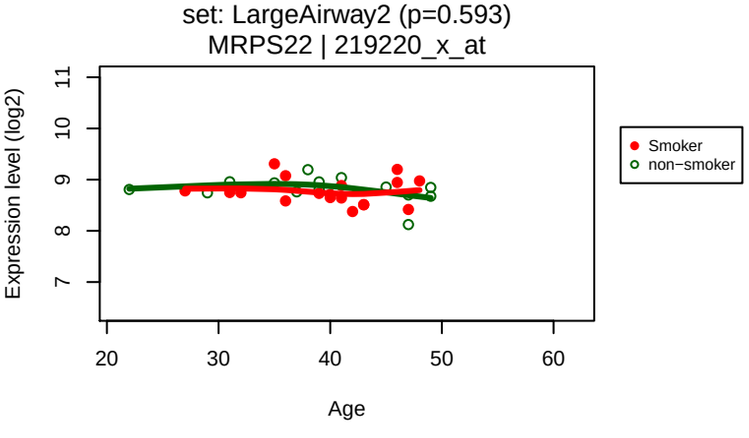

Gene expression ratio (smokers/nonsmokers)
gene: MRPS22 | 219220_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0788
p=0.4441
p=0.0221
p=0.0139
p=0.593
p=0.0146
p=0.5845
p=0.1169
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

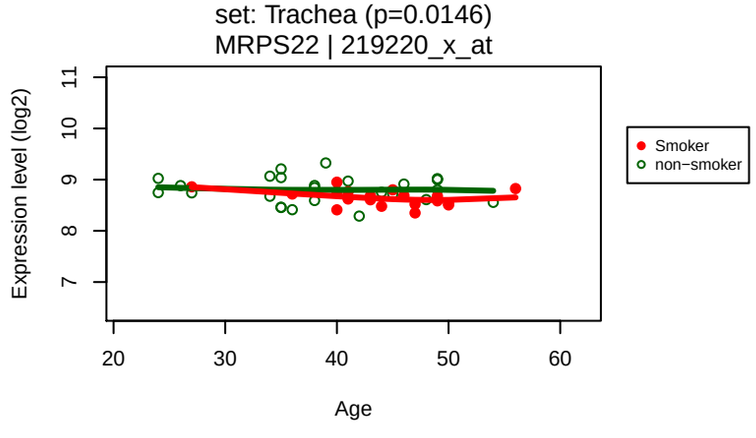

Gene expression ratio in Males
gene: MRPS22 | 219220_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8574
p=0.9168
p=0.0251
p=0.6343
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPS22 | 219220_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5348
p=0.2761
p=0.136
p=0.1256
SmallAirway2
LargeAirway2
Trachea
Macrophage2

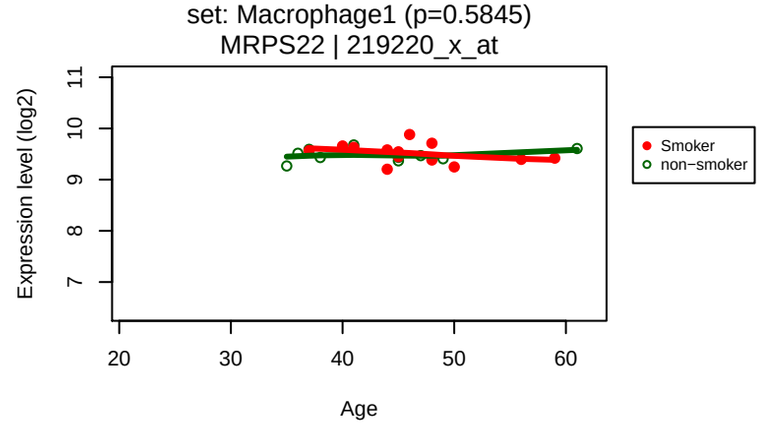

set: SmallAirway1 (p=0.0788)
MRPS22 | 219220_x_at
Expression level (log2)
Age
Smoker
non−smoker

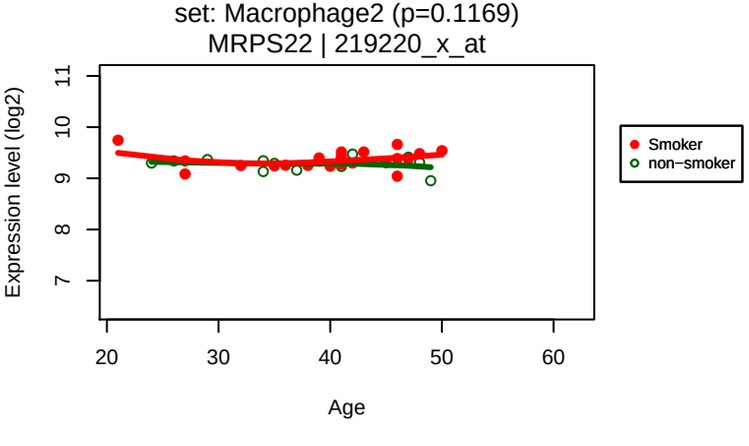

set: SmallAirway2 (p=0.4441)
MRPS22 | 219220_x_at
Expression level (log2)
Age
Smoker
non−smoker

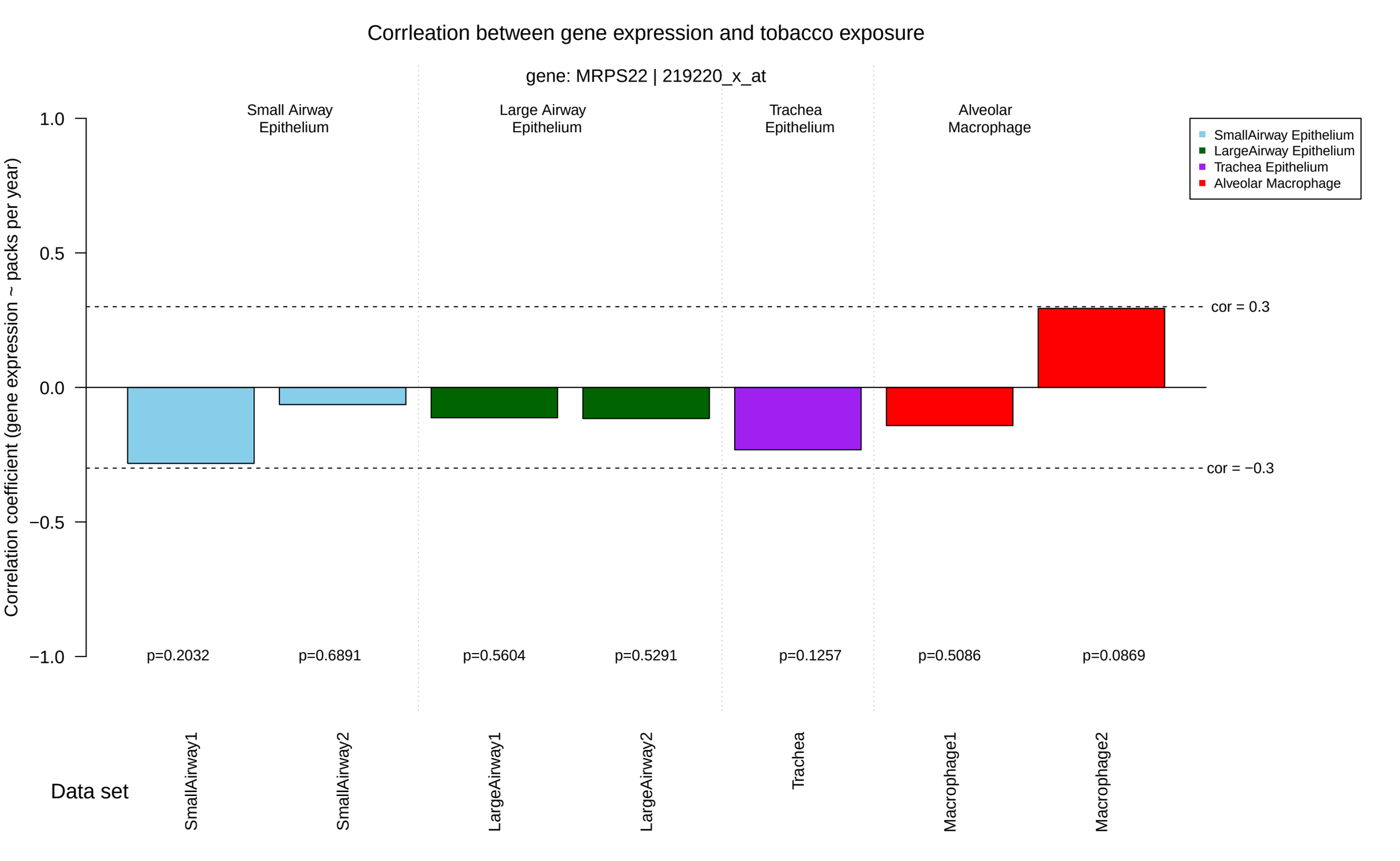

set: LargeAirway1 (p=0.0139)
MRPS22 | 219220_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.593)
MRPS22 | 219220_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0146)
MRPS22 | 219220_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5845)
MRPS22 | 219220_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1169)
MRPS22 | 219220_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPS22 | 219220_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2032
p=0.6891
p=0.5604
p=0.5291
p=0.1257
p=0.5086
p=0.0869
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2